Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If microwaves, X rays, infrared, gamma rays, ultra-violet, radio waves and visible parts of the electromagnetic spectrum are denoted by $\mathrm{M}, \mathrm{X}, \mathrm{I}, \mathrm{G}, \mathrm{U}, \mathrm{R}$ and V then which of the following is the arrangement in ascending order of wavelength ?

(A) $\mathrm{R}, \mathrm{M}, \mathrm{I}, \mathrm{V}, \mathrm{U}, \mathrm{X}$ and G

(B) $\mathrm{M}, \mathrm{R}, \mathrm{V}, \mathrm{X}, \mathrm{U}, \mathrm{G}$ and I

(C) $\mathrm{G}, \mathrm{X}, \mathrm{U}, \mathrm{V}, \mathrm{I}, \mathrm{M}$ and R

(D) I, M, R, U, V, X and G

Q.2

Which of the following electromagnetic waves has minimum frequency ?

(A) Microwaves

(B) Audible waves

(C) Ultrasonic wave

(D) Radiowaves

Q.3

All components of the electromagnetic spectrum in vacuum have the same

(A) energy

(B) velocity

(C) wavelength

(D) frequency

Q.4

Frequency of a wave is 6×10^{15} Hz. The wave is

(A) radiowave

(B) microwave

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(C) x-ray

(D) None of these

Q.5

A lamp emits monochromatic green light uniformly in all directions. The lamp is 3% efficient in converting electrical power to electromagnetic waves and consumes $100\ \mathrm{W}$ of power. Then what will be the amplitude of the electric field (in $\mathrm{V/m}$) associated with the electromagnetic radiation at a distance of $5\ \mathrm{m}$ from the lamp?

(A) 2.68

(B) 2.78

(C) 1.68

(D) 1.78

Q.6

For wave propagation of a $10\mathrm{MHz}$ signal, what should be the maximum electron density in ionosphere

(A) $\sim 1.2 \times 10^{12}\ \mathrm{m}^{-3}$

(B) $\sim 10^{6}\ \mathrm{m}^{-3}$

(C) $\sim 10^{14}\ \mathrm{m}^{-3}$

(D) $\sim 10^{22}\ \mathrm{m}^{-3}$

Q.7

The energy density of electromagnetic wave in vacuum is given by the relation

(A) $\frac{E^2}{2\varepsilon_0}+\frac{B^2}{2\mu_0}$

(B) $\frac{1}{2}\varepsilon_0 \mathrm{E}^2+\frac{1}{2}\mu_0\ \mathrm{B}^2$

(C) $\frac{\mathrm{E}^2+\mathrm{B}^2}{\mathrm{c}}$

(D) $\frac{1}{2}\varepsilon_0 \mathrm{E}^2+\frac{\mathrm{B}^2}{2\mu_0}$

Q.8

The waves which are electromagnetic in nature are

(A) sound waves and light waves

(B) water waves and radio waves

(C) light waves and X-rays

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(D) sound waves and water waves

Q.9

Electromagnetic radiation of highest frequency is

(A) infrared radiations

(B) visible radiation

(C) radio waves

(D) γ -rays

Q.10

In which one of the following regions of the electromagnetic spectrum will the vibrational motion of molecules give rise to absorption?

(A) Ultraviolet

(B) Microwaves

(C) Infrared

(D) Radio waves

Q.11

Which of the following statements is true?

(A) The wavelength of microwaves is greater than that of UV-rays

(B) The wavelength of IR rays is lesser than that of UV-rays

(C) The wavelength of microwaves is lesser than that of IR rays

(D) Gamma rays has largest wavelength in the electromagnetic spectrum.

Q.12

Which of the following electromagnetic waves has minimum frequency?

(A) Microwaves

(B) Audible waves

(C) Ultrasonic wave

(D) Radiowaves

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Q.13

The range of wavelength of the visible light is

(A) $10\mathring{A}$ to $100\mathring{A}$

(B) $4,000\mathring{A}$ to $8,000\mathring{A}$

(C) $8,000\mathring{A}$ to $10,000\mathring{A}$

(D) $10,000\mathring{A}$ to $15,000\mathring{A}$

Q.14

Which of the following shows green house effect?

(A) Ultraviolet rays

(B) Infrared rays

(C) X-rays

(D) None of these

Q.15

Given below is a list of E.M spectrum and its use. Which one does not match?

(A) U.V. rays - finger prints detection

(B) I.R.. rays - Secret writing on ancient walls

(C) X - rays - Atomic structure

(D) Microwaves - forged document detection

Q.16

If v_s, v_x and v_m are the speed of soft gamma rays, X-rays and microwaves respectively in vacuum, then

(A) $\mathrm{v_s} > \mathrm{v_x} > \mathrm{v_m}$

(B) $\mathrm{v_s} < \mathrm{v_x} < \mathrm{v_m}$

(C) $v_s > v_x < v_m$

(D) $\mathrm{v_s} = \mathrm{v_x} = \mathrm{v_m}$

Q.17

Given below is a list of electromagnetic spectrum and its mode of production. Which one does not match?

(A) Gamma rays - Radioactive decay of the nucleus

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(B) Ultraviolet rays - Magnetron valve

(C) Radio wave - Rapid acceleration and deceleration of electrons in conducting wires

(D) X-rays - coolidge tube

Q.18

A metal block is exposed to beams of X-ray of different wavelengths X-rays of which wavelength penetrate most

(A) $2\mathring{A}$

(B) $4\mathring{A}$

(C) $6\mathring{A}$

(D) $8\mathring{A}$

Q.19

In radio therapy. X-rays are used to

(A) Detect bone fractures

(B) Treat cancer by controlled exposure

(C) Detect heart diseases

(D) Detect fault in radio receiving circutts

Q.20

The potential difference applied to an X -ray tube is increased. As a result, in the emitted radiation

(A) The intensity increases

(B) The minimum wavelength increases

(C) The intensity decreases

(D) The minimum wavelength decreases

Answers & Solutions

Q.1 Answer:

G, X, U, V, I, M and R

Solution:

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Gamma rays $<$ X-rays $<$ Ultra violet $<$ Visible rays $<$ Infrared rays $<$ Microwaves $<$ Radio waves.

Q.2 Answer:

Radiowaves

Solution:

$\lambda \alpha \frac{1}{v}$

Wave length of RW are highest, so frequency is lowest.

Q.3 Answer:

velocity

Solution:

All components of electromagnetic spectrum travel in vacuum with velocity $3 \times 10^{8}\ \mathrm{m/s}$

Q.4 Answer:

None of these

Solution:

Wave is UV rays.

Q.5 Answer:

2.68

Solution:

Wavelength of monochromatic green light $= 5.5 \times 10^{-5}\ \mathrm{cm}$

$$\text{Intensity I} = \frac{\text{Power}}{\text{Area}}$$
$$= \frac{100 \times (3/100)}{4\pi(5)^2} = \frac{3}{100\pi}\mathrm{Wm}^{-2}$$

Now, half of this intensity (I) belongs to electric field and half of that to magnetic field,

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therefore,

$$\frac{\mathrm{I}}{2}=\frac{1}{4} \varepsilon_0 \mathrm{E}_0^2 \mathrm{C}$$

$$\text{or } \mathrm{E}_0=\sqrt{\frac{2 \mathrm{I}}{\varepsilon_0 \mathrm{C}}}$$

$$=\sqrt{\frac{2 \times\left(\frac{3}{100} \pi\right)}{\left(\frac{1}{4 \pi \times 9 \times 10^9}\right) \times\left(3 \times 10^8\right)}}=\sqrt{\frac{6}{25} \times 30}=\sqrt{7.2}$$

$$\therefore \quad \mathrm{E}_0=2.68 \mathrm{~V} / \mathrm{m}$$

Q.6 Answer:

$\sim 1.2 \times 10^{12} \mathrm{~m}^{-3}$

Solution:

If maximum electron density of the ionosphere is $\mathrm{N}_{\max}$ per m^3 then the critical frequency f_{c} is given by $\mathrm{f}_{\mathrm{c}}=9\left(\mathrm{~N}_{\max }\right)^{1 / 2}$

$\Rightarrow 10 \times 10^6=9(N)^{1 / 2} \Rightarrow N=1.2 \times 10^{12} \mathrm{~m}^{-3}$

Q.7 Answer:

$\frac{1}{2} \varepsilon_0 \mathrm{E}^2+\frac{\mathrm{B}^2}{2 \mu_0}$

Solution:

No solution available

Q.8 Answer:

light waves and X-rays

Solution:

Light waves are electromagnetic waves.

Q.9 Answer:

γ -rays

Solution:

$v_{\gamma-\text{ rays}}>v_{\text{visible radiation}}>v_{\text{infrared}}>v_{\text{Radio waves}}$

Q.10 Answer:

Microwaves

Solution:

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Molecular spectra due to vibrational motion lie in the microwave region of EM-spectrum. Due to Kirchhoff's law in spectroscopy the same will be absorbed.

Q.11 Answer:

The wavelength of microwaves is greater than that of UV-rays

Solution:

Wavelength order

G < X < U < V < I < M < R

Q.12 Answer:

Radiowaves

Solution:

$\lambda \alpha \frac{1}{v}$

Radio wave has minimum frequency

Q.13 Answer:

$4,000\mathring{A}$ to $8,000\mathring{A}$

Solution:

Wavelength of visible spectrum is $3900\mathring{A} - 7800\mathring{A}$

Q.14 Answer:

Infrared rays

Solution:

Infrared radiations reflected by low lying clouds and keeps the earth warm.

Q.15 Answer:

Microwaves - forged document detection

Solution:

Ultravoilet radiations are used in the detection of invisible writing, forged documents, finger prints in forensic lab. While microwaves are used in microwave oven.

Q.16 Answer:

$\mathrm{v_s} = \mathrm{v_x} = \mathrm{v_m}$

Solution:

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No solution available

Q.17 Answer:

Ultraviolet rays - Magnetron valve

Solution:

UV-rays - Magnetron is not correct.

Q.18 Answer:

$2\mathring{A}$

Solution:

Penetrating power is greater for lower wavelength.

Q.19 Answer:

Treat cancer by controlled exposure

Solution:

Radiation therapy uses high-energy particles or waves, such as x-rays, gamma rays, electron beams, or protons, to destroy or damage cancer cells.

Q.20 Answer:

The minimum wavelength decreases

Solution:

$\lambda_{\min} = \frac{hc}{eV}$ **or** $\lambda_{nin} \propto \frac{1}{V}$ **On increasing potential.** λ_{mi} **decreases.**

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